

1. Write a program to reverse a word using loop? (Not to use inbuilt functions)
Sample Input:
String: TEMPLE
Sample Output:
Reverse String: ELPMET
2. Write a program to convert the given string to integer?
Sample Input:
String: 1234
Sample Output:
Out put String: 1234
3. Write a program to check the entered user name is valid or not. Get both the inputs from the user.
Sample Input:
Enter the user name: Saveetha@789
Reenter the user name: Saveetha@123
Sample Output:
User name is Invalid
4. Write a program that would sort a list of names in alphabetical order Ascending or Descending, choice get from the user?
5. Write a program to print the special characters separately and print number of Special characters in the line?
6. Write a program to print the number of vowels in the given statement?
Sample Input:
Saveetha School of Engineering
Sample Output:
Number o vowels = 12
7. Write a program to print consonants and vowels separately in the given word
Sample Input:
Given Word: Engineering
Sample Output:
Consonants: n g n r n g
Vowels: e i e ei
8. Write a program that finds whether a given character is present in a string or not. In case it is present it prints the index at which it is present. Do not use built-in find functions to search the character.
Sample Input:
Enter the string: I am a programmer
Enter the character to be searched: p

Sample Output:

P is found in string at index: 8

Note: Check for non available Character in the given statement as Hidden Test case.

9. Write a program that accepts a string from user and displays the same string after removing vowels from it.

Sample Input & Output:

Enter a string: we can play the game

The string without vowels is: w cn ply thgm

Arrays:

10. Write a program for matrix multiplication?

Sample Input:

Mat1 = 1 2
 5 3

Mat2 = 2 3
 4 1

Sample Output:

Mat Sum = 10 5
 22 18

11. Write a program for Merge two sorted arrays using Array list

Input: arr1[] = { 1, 3, 4, 5}, arr2[] = {2, 4, 6, 8}

Output: arr3[] = {1, 2, 3, 4, 4, 5, 6, 8}

12. Find the Mean, Median, Mode of the array of numbers?

Sample Input::

Array of elements = {16, 18, 27, 16, 23, 21, 19}

Sample Output:

Mean = 20

Median = 19

Mode = 16

13. Write a program to find the number of composite numbers in an array of elements

Sample Input::

Array of elements = {16, 18, 27, 16, 23, 21, 19}

Sample Output:

Number of Composite Numbers = 5

Patterns :

14. Write a program to print Right Triangle Star Pattern

Sample Input:: n = 5

Output:

```
  *
 * *
* * *
* * * *
```

* * * * *

15. Write a program to print the below pattern?

```

          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1

```

16. Write a program to print rectangle symbol pattern.

Get the symbol as input from user

17. Write a program to print the following pattern

Sample Input:

Enter the number to be printed: 1

Max Number of time printed: 3

```

1
11
111
11
1

```

18. Write a program to print the Inverted Full Pyramid pattern?

19. Write a program to print the following pattern

Sample Input:

Enter the Character to be printed: %

Max Number of time printed: 3

```

%
% %
% % %

```

20. Write a program to print hollow square symbol pattern?

21. Write a program to print the below pattern

```
1
2 2
3 3 3
4 4 4 4
```

22. Write a program to print the below pattern

```
1
4 9
16 25 36
49 64 81 100
```

23. Write a program to print the below pattern

```
1
2 2
3 3 3
4 4 4 4
3 3 3
2 2
1
```

24. Write a program to print inverted pyramid pattern.

Input: no of rows: 3

Output

```
*****
***
*
```

General:

25. Write a program to reverse a number using loop?(Get the input from user)

Sample Input:

Number: 14567

Sample Output:

Reverse Number: 76541

26. Write a program to convert the given decimal to binary and print the reverse of the binary decimal.

Input: 11

Output: 13

Explanation: $(11)_{10} = (1011)_2$.

After reversing the bits we get:

$(1101)_2 = (13)_{10}$.

27. Write a program to find whether the person is eligible for vote or not. And if that particular person is not eligible, then print how many years are left to be eligible.

28. Find the LCM and GCD of n numbers?

29. Write a program using function to calculate the simple interest. Suppose the customer is a senior citizen. He is being offered 12 percent rate of interest; for all other customers, the ROI is 10 percent.

Sample Input:

Enter the principal amount: 200000

Enter the no of years: 3

Is customer senior citizen (y/n): n

Sample Output:

Interest: 60000

30. Write a program to print the Fibonacci series.

Sample Input:

Enter the n value: 6

Sample Output:

0 1 1 2 3 5

31. Java Program to Find Even Sum of Fibonacci Series Till number N?

Sample Input: n = 4

Sample Output: 33

(N = 4, So here the fibonacci series will be produced from 0th term till 8th term: 0, 1, 1, 2, 3, 5, 8, 13, 21)

Sum of numbers at even indexes = $0 + 1 + 3 + 8 + 21 = 33$)

32. Write a program to print the numbers from M to N by skipping K numbers in between?

Sample Input:

M = 50

N = 100

K = 7

Sample Output:

50, 58, 66, 74,

33. Write a program to print all the composite numbers between a and b?

Sample Input:

A = 12

B = 19

Sample Output

14, 15, 16, 18

34. Find the factorial of n?

Sample Input:

N = 4

Sample Output:

4 Factorial = 24

35. Find the year of the given date is leap year or not

Sample Input:

Enter Date: 04/11/1947

Sample Output:

Given year is Non Leap Year

36. Write a program to print the given number is Perfect number or not?

Sample Input:

Given Number: 6

Sample Output:

It's a Perfect Number

37. Write a program to find the square, cube of the given decimal number

Sample Input:

Given Number: 0.6

Sample Output:

Square Number: 0.36

Cube Number:0.216

38. Find the n^{th} odd number after n odd number

Sample Input: N : 7

Sample Output:

Hence the values printed for i are 1 , 3 , 5.

40 Program to find the frequency of each element in the array.

Sample Input & Output:

{1, 2, 8, 3, 2, 2, 2, 5, 1}

Pseudo:

Element Frequency	

1	2
2	4
8	1
3	1
4	1

39. Program to find whether the given number is Armstrong number or not

Sample Input:

Enter number: 153

Sample Output:

Given number is Armstrong number

40. Write a program to find the sum of digits of N digit number (sum should be single digit)

Sample Input:

Enter N value: 3

Enter 3 digit numbers: 143

41. Write a program to find the square root of a perfect square number(print both the positive and negative values)

Sample Input:

Enter the number: 6561

Sample Output:

Square Root: 81, -81

42. Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:

Enter * to exit...

Enter any character: W

Enter any character: d

Enter any character: A

Enter any character: G

Enter any character: g

Enter any character: H

Enter any character: *

Sample Output:

Total count of lower case:2

43. Write a program to print the first n perfect numbers. (Hint Perfect number means a **positive integer that is equal to the sum of its proper divisors**)

Sample Input:

N = 3

Sample Output:

First 3 perfect numbers are: 6 , 28 , 496

44. Write a program to convert Decimal number equivalent to Binary number and octal numbers?

Sample Input:

Decimal Number: 15

Sample Output:

Binary Number = 1111

Octal = 17

45. Write a program using choice to check

Case 1: Given string is palindrome or not

Case 2: Given number is palindrome or not

Sample Input:

Case = 1

String = MADAM

Sample Output:

Palindrome

46. Find the Mth maximum number and Nth minimum number in an array and then find the sum of it and difference of it.

Sample Input:

Array of elements = {14, 16, 87, 36, 25, 89, 34}

M = 1

N = 3

Sample Output:

1stMaximum Number = 89

3rdMinimum Number = 25

Sum = 114

Difference = 64

47. Write a Program to Find the Nth Largest Number in a array.

Sample Input:

List : {14, 67, 48, 23, 5, 62}

N = 4

Sample Output:

4th Largest number: 23

48. Write a Program to Remove the Duplicate Items from a array.

49. Bank is a class that provides method to get the rate of interest. But, rate of interest may differ according to banks. For example, SBI, ICICI and AXIS banks are providing 8.4%, 7.3% and 9.7% rate of interest. Write a Java program for above scenario.

Sample Input SBI, 8.4.

50. Bring out the situation in which member names of a subclass hide members by the same name in the super class. How it can be resolved? Write Suitable code in Java and Implement above scenario with the Parametrized Constructor (accept int type parameter) of the Super Class can be called from Sub Class Using super () and display the input values provided.